

Milestone 5: Business Logic Implementation

Objective

The objective of this phase was to implement business rules and automation logic required for processing leave requests efficiently. Business logic was developed to automate calculations, status updates, and AI-based decision support, thereby reducing manual intervention and improving system accuracy.

Overview

After designing the Leave Request form, business logic was implemented to automate key operations within the Smart Leave Approval System. The automation ensures that leave requests are processed consistently and that managers receive relevant information for decision-making.

The implementation was carried out using ServiceNow Business Rules, Client Scripts, and Flow Designer.

Business Logic Components

1. Leave Number Generation

Description

Each leave request is assigned a unique identification number automatically when a record is created.

Purpose

- >Ensures uniqueness of records.
- >Simplifies tracking and reporting.
- >Improves record management.

Example

LEV000001
LEV000002

2. Total Days Calculation

Description

The system automatically calculates the total number of leave days based on the selected start date and end date.

Formula

Total Days = End Date - Start Date + 1

Example

Start Date	End Date	Total Days
01-Jun-2026	03-Jun-2026	3

Benefits

1. Eliminates manual calculations.

- 2.Reduces errors.
- 3.Improves accuracy.

3. Approval Status Management

Description

The system automatically manages the leave request status throughout the approval lifecycle.

Status Values

Status	Description
Pending	Waiting for manager approval
Approved	Request approved
Rejected	Request rejected

The screenshot shows the 'Notification [sysevent_email_action]' configuration page in ServiceNow. The page title is 'New record'. A blue banner at the top explains the purpose of notifications: 'Use Notifications to notify users about specific activities in ServiceNow, such as updates to incidents or change requests. Notifications allow administrators to specify'.

Configuration details include:

- Name:** Leave Approved
- Table:** Leave Request [x_1877519_smart_0_leave...]
- Application:** Smart Leave Approval System
- Active:** ☒
- Mandatory:** ☐
- Category:** Uncategorized

A note states: 'A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.' Below this is the 'Allow Digest' checkbox, which is currently unchecked.

The 'Who will receive' tab is selected under the 'When to send' section. A blue banner states: 'Notifications can be sent to specific Users and Groups or to User/Groups in fields on the record that generated this notification.'

Configuration for recipients includes:

- Users:**
- Groups:**
- Users/Groups in fields:** Employee
- Subscribable:** ☐

At the bottom, there is a 'Submit' button and a 'Related Links' section with a link to 'Advanced view'.

Process

1. Employee submits leave request.
2. Status automatically becomes **Pending**.
3. Manager reviews request.

Status changes to **Approved** or **Rejected**.

Notification [ysevent_email_action]
New record

Name: Leave Approved

Application: Smart Leave Approval System

* Table: Leave Request [x_1877519_smart_0_leav...]

Mandatory: ☐

* Category: Uncategorized

A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.

Allow Digest: ☐

When to send | Who will receive | What it will contain

When using an Email Template, the Subject and Message content is picked from that template unless a different Subject or Message content is specified in this form.

Email template: Unsubscribe and Preferences

Subject: Your Leave Request Has Been Approved

Message HTML:

B I U ↶ ↷ Verdana 8pt
 Hello \${employee},
 Your leave request has been approved.
 Leave Type: \${leave_type}
 Start Date: \${start_date}
 End Date: \${end_date}
 Status: Approved
 Thank You

Notification [ysevent_email_action]
New record

Name: Leave Rejected

Application: Smart Leave Approval System

* Table: Leave Request [x_1877519_smart_0_leav...]

Mandatory: ☐

* Category: Uncategorized

A digest combines and summarises multiple updates for a single or multiple target records within a set time interval.

Allow Digest: ☐

When to send | Who will receive | What it will contain

Notifications can be sent (if the specified Conditions are met) under one of the following circumstances:

- A record is Inserted or Updated into the Table specified above
- The specified event is fired
- Via a Flow Action

Send when: generation_type | Record inserted or updated | engine

Updated: action_update

Inserted: action_insert

Conditions: condition

Add Filter Condition Add OR Clause

Status is Rejected | rejected

Submit

Related Links
Advanced view

4. AI Risk Assessment Logic

Description

A rule-based AI mechanism was implemented to evaluate leave requests based on leave duration.

Risk Classification

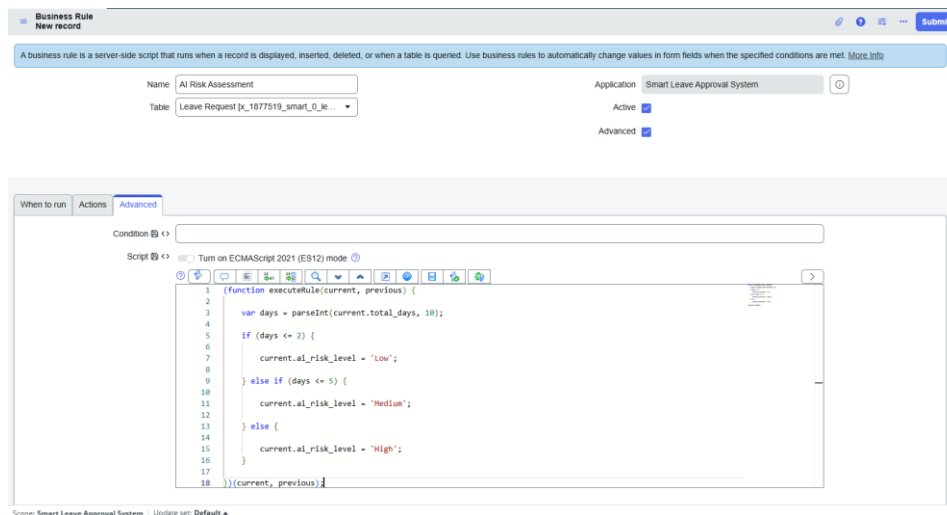
Total Days	Risk Level
1 – 3 Days	Low Risk
4 – 7 Days	Medium Risk
More than 7 Days	High Risk

Purpose

Helps managers prioritize reviews.

Highlights potentially critical leave requests.

Supports decision-making.



5. AI Recommendation Logic

Description

The system generates recommendations based on leave duration and risk level.

Example Recommendations

Low Risk

Leave request appears normal.

Medium Risk

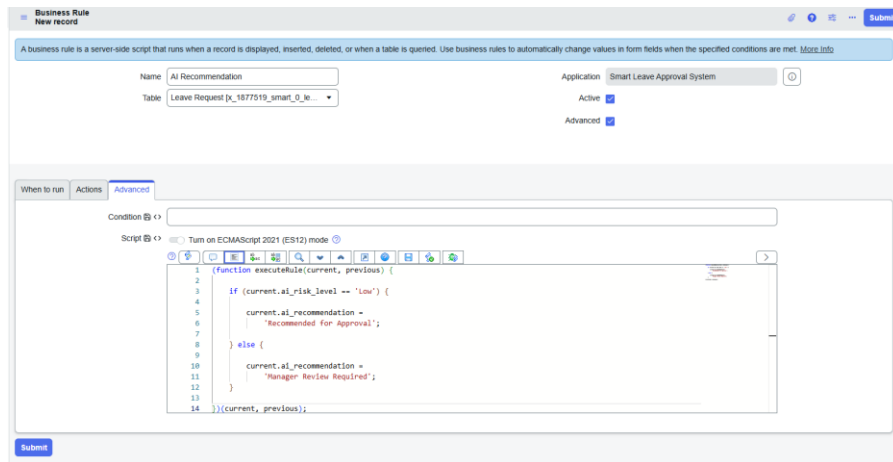
Manager review recommended.

High Risk

Extended leave requires additional review.

Benefits

- ✓ Provides decision support.
- ✓ Ensures consistent evaluation.
- ✓ Reduces approval uncertainty.



6. Data Validation Rules

Description

Validation logic ensures that users enter complete and accurate information before submitting a leave request.

Validation Checks

Mandatory Fields

Employee

Leave Type

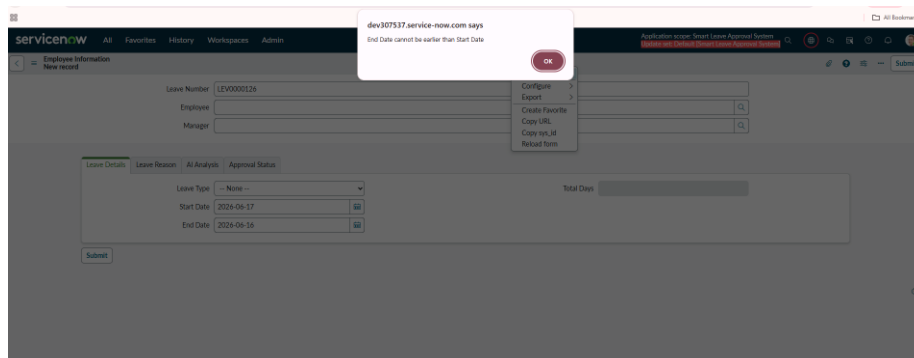
Start Date

End Date

Leave Reason

Date Validation

End Date cannot be earlier than Start Date.



7. Record Creation Automation

Description

When an employee submits a leave request, the system automatically:

1. Creates a new leave record.
2. Assigns a unique leave number.
3. Calculates total leave days.
4. Generates AI risk assessment.
5. Generates AI recommendation.
6. Sets approval status to Pending.

Result

All essential information is automatically populated before manager review.

Workflow Integration

The business logic works together with the approval workflow to ensure smooth processing of leave requests.

Workflow Sequence

Employee Submits Request



Calculate Total Days



Generate AI Risk Assessment



Generate AI Recommendation



Set Status to Pending



Send for Manager Approval